Nishal Stanislaus Silva, PhD

Machine Learning Research Engineer | Deep Learning, Time-Series, Pattern Detection | Applied ML
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Summary

Machine Learning Engineer with 10+ years across industry and academia. I build end-to-end ML systems (data ingestion, feature engineering, training, deployment, monitoring), focused on real-time inference, time-series, pattern/anomaly detection, and computer vision. Deployed results include <30ms inference at >90% F1 and up to 300% efficiency gains.

Skills

ML & AI: PyTorch, TensorFlow/Keras, scikit-learn, HuggingFace, OpenCV, LLMs, RAG, Generative AI (VAE, Diffusion)
Specialties: Time-Series, Real-Time Inference, Anomaly & Pattern Detection, Computer Vision, DSP, Multimodal Data
MLOps & Data: AWS (EC2, S3, ECS, SageMaker), Docker, Kubernetes, MLflow, Airflow, CI/CD, ONNX, Snowflake, SQL
Languages & Tools: Python, C++, C, JavaScript, MATLAB, Git, Linux, Jenkins, Pytest, REST APIs, NumPy, Pandas

Experience

Open Source / MUSMET external collaborator Toronto, ON
Independent Researcher *Jan 2026 - Present*
- Released Nebula, an open-source C++17 time-series feature-extraction library with ARM latency benchmarks.
- Fine-tuning a generative foundation model (LoRA, multi-GPU) to synthesize training data for limited-label regimes.
- Benchmarked DTW, Gaussian and Wav2Vec2 scoring to detect subtle speech deviations; 100k trials, ROC-AUC 0.822.
- Authored 3 papers on embedded real-time pattern detection and edge-to-server distributed feature extraction.

University of Trento Trento, Italy
Postdoctoral Researcher *Jan 2025 - Dec 2025*
- Owned EU-funded ML pipelines (MUSMET) for streaming sensor data: DSP, training, edge deployment, monitoring.
- Built real-time inference services hitting <30ms latency at >90% F1 on edge devices, with reproducible pipelines.
- Designed benchmark suites (baselines + ablations) to quantify architecture/feature trade-offs under compute limits.
- Developed a real-time pipeline synchronizing audio, EEG, and mixed-reality tracking streams across 4 concurrent users.

McGill University Montreal, QC
Visiting Researcher *May 2024 - Aug 2024*
- Prototyped real-time gesture detection fusing IMU and audio on embedded hardware (F1 0.78), with compute profiling.
- Explored diffusion- and VAE-based synthetic data generation to improve model robustness with limited labels.

Forestpin (Pvt) Ltd Colombo, Sri Lanka
Technical Consultant *Aug 2020 - Feb 2021*
- Designed and deployed ML and statistical pipelines for financial forensics and risk detection on enterprise data.
- Built SQL + Python anomaly-scoring services; turned compliance requirements into technical detection specs.

MAS Holdings (Pvt) Ltd Colombo, Sri Lanka
Research Engineer *Jan 2015 - Jul 2020*
- Led end-to-end development of ML and computer vision systems for manufacturing at scale; up to 300% efficiency gains.
- Shipped automated visual inspection (OpenCV, PLC reject, human-in-the-loop review), cutting inspection time by 99.5%.
- Designed the alignment algorithm and printer integration for Promptly, an on-demand printing service now in 4 countries.
- Built real-time ingestion and ETL pipelines for high-frequency IoT time-series data across distributed sites.
- Mentored engineers and established code review and CI/CD practices to accelerate ML adoption across teams.

Education

Ph.D., Information Engineering and Computer Science, University of Trento, Italy *Jan 2025*
M.Sc., Telecommunication and Electronic Engineering, Sheffield Hallam University, UK *Aug 2018*
B.Eng. (Hons), Electronic Engineering, Sheffield Hallam University, UK *Mar 2014*

Publications & Awards

- **Publications:** 10+ peer-reviewed papers (JAES, IEEE, ACM) and 4 open benchmark datasets; nishal.xyz/publications
- **Awards:** MIDI Innovation Awards finalist (2023); GMOA (Sri Lanka) recognition for COVID-19 vitals monitoring.